

UML Package

B.Tech CSE 2025–29 · Software Engineering & Project Management · Semester III

Dashboard basis: The project author has developed three dashboards—Student, Faculty and Admin. This document uses the supplied walkthrough to describe them and distinguishes those dashboard details from requirements, integrations or metrics that are proposed or not independently verified.

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1. Design rationale: cohesion and loose coupling

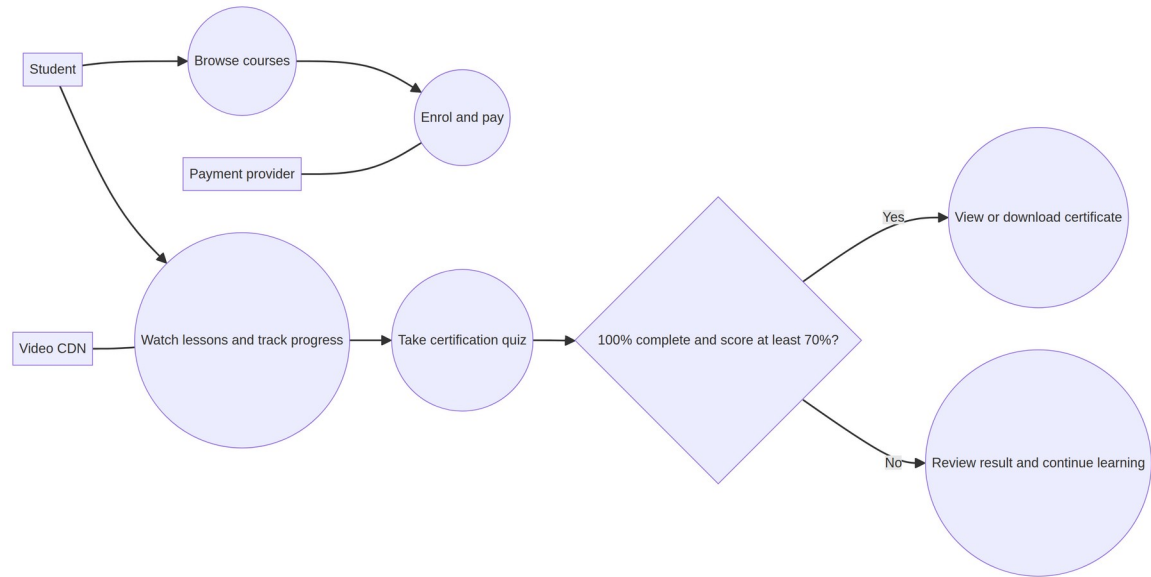
Each module has one clear responsibility: Identity authenticates users and enforces roles; Course Catalogue owns course and lesson metadata; Enrolment records a learner's place on a course; Payment Adapter communicates with a payment provider; Learning Progress records completion; Assessment evaluates quiz attempts; Certificate Service issues an award only after eligibility is confirmed; Production Tracking records faculty/editor progress; and Reporting assembles dashboard views.

Modules communicate through narrow interfaces and stable identifiers. In particular, video bytes are delivered by a CDN rather than streamed through the application API, and payment outcomes are received through a provider adapter rather than embedding provider logic in enrolment rules. This limits the effect of replacing a CDN or payment provider. The Certificate Service receives a verified eligibility result; it does not make eligibility decisions from client-side UI state. Firestore/local JSON coexistence described in the dashboard notes is an implementation detail that needs reconciliation logic; the logical design expects a single authoritative record per transaction.

2. Use-case diagrams by dashboard

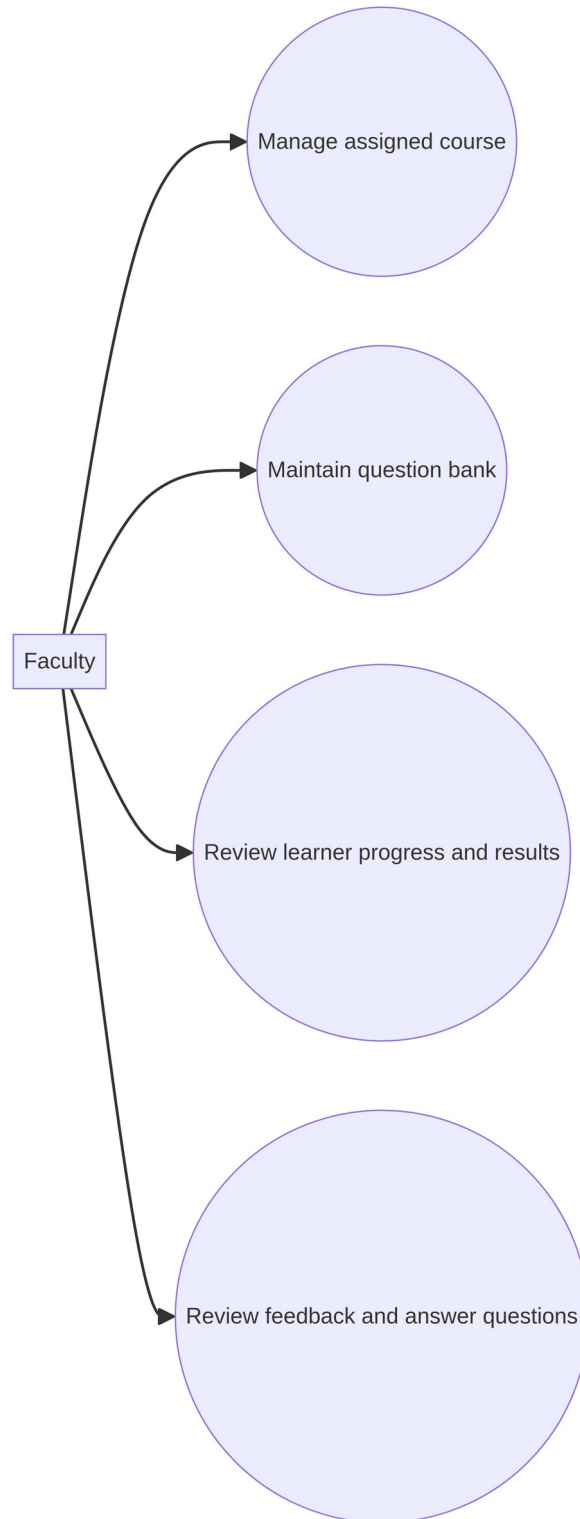
The use cases are grouped by the three implemented dashboards so each role's responsibilities are easy to distinguish.

2.1 Student dashboard



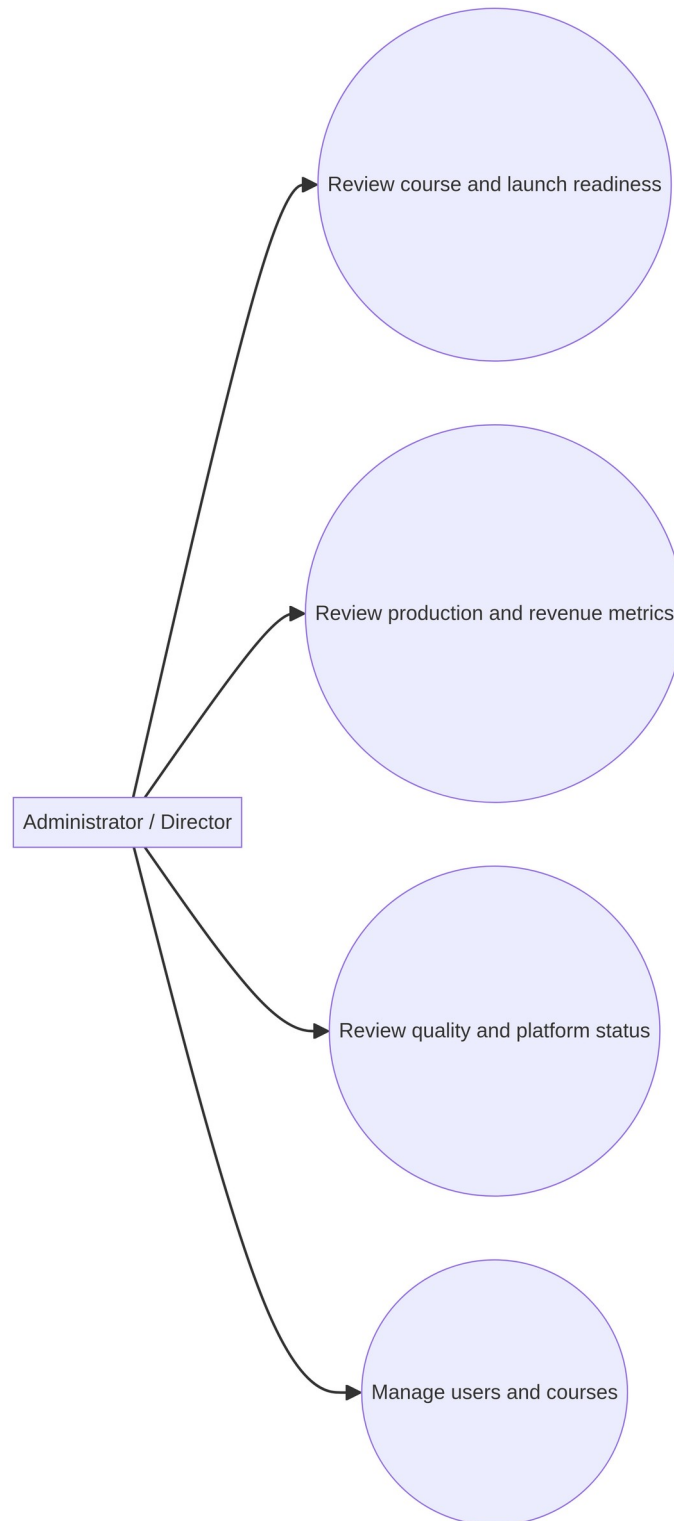
Student dashboard use cases

2.2 Faculty dashboard



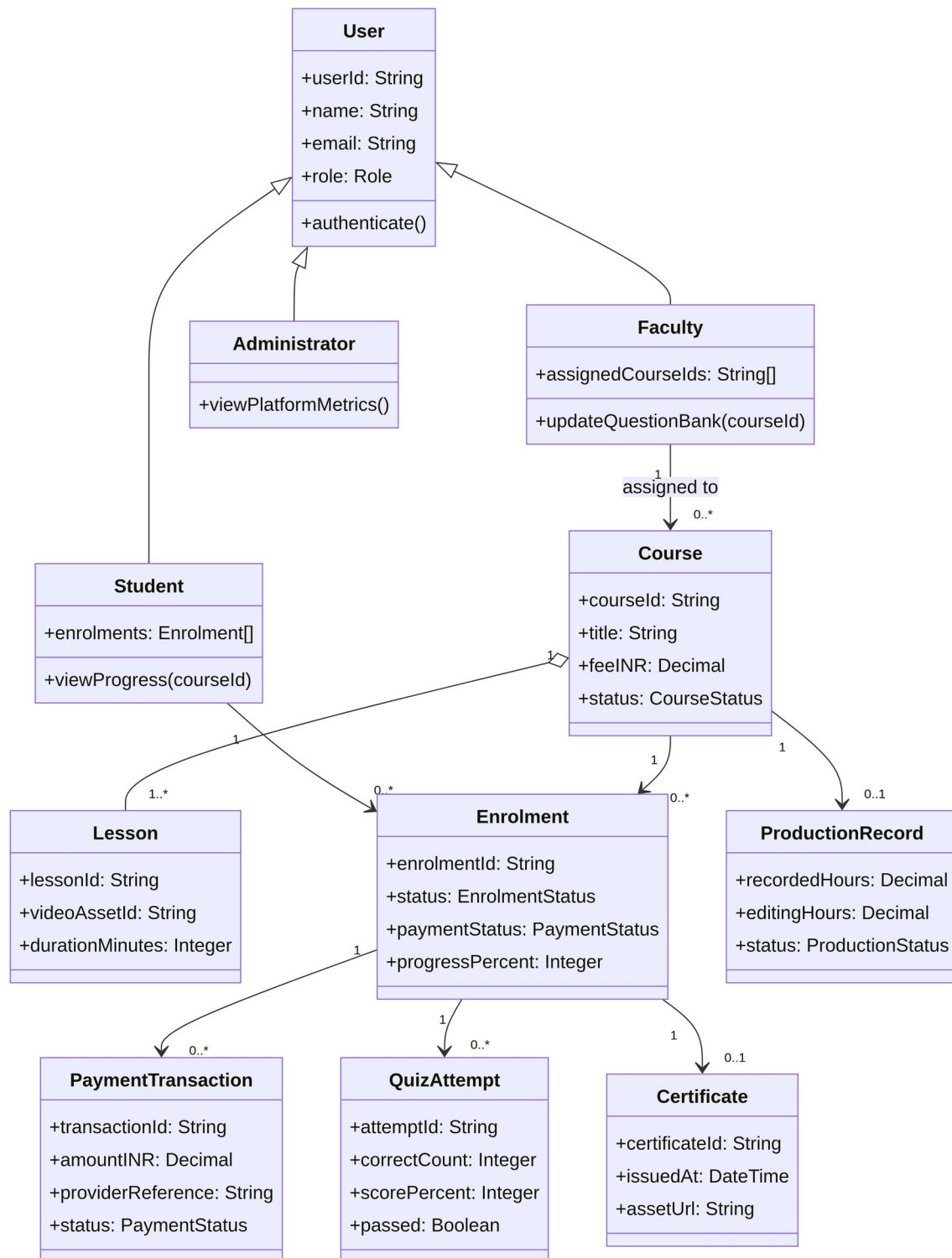
Faculty dashboard use cases

2.3 Administrator dashboard



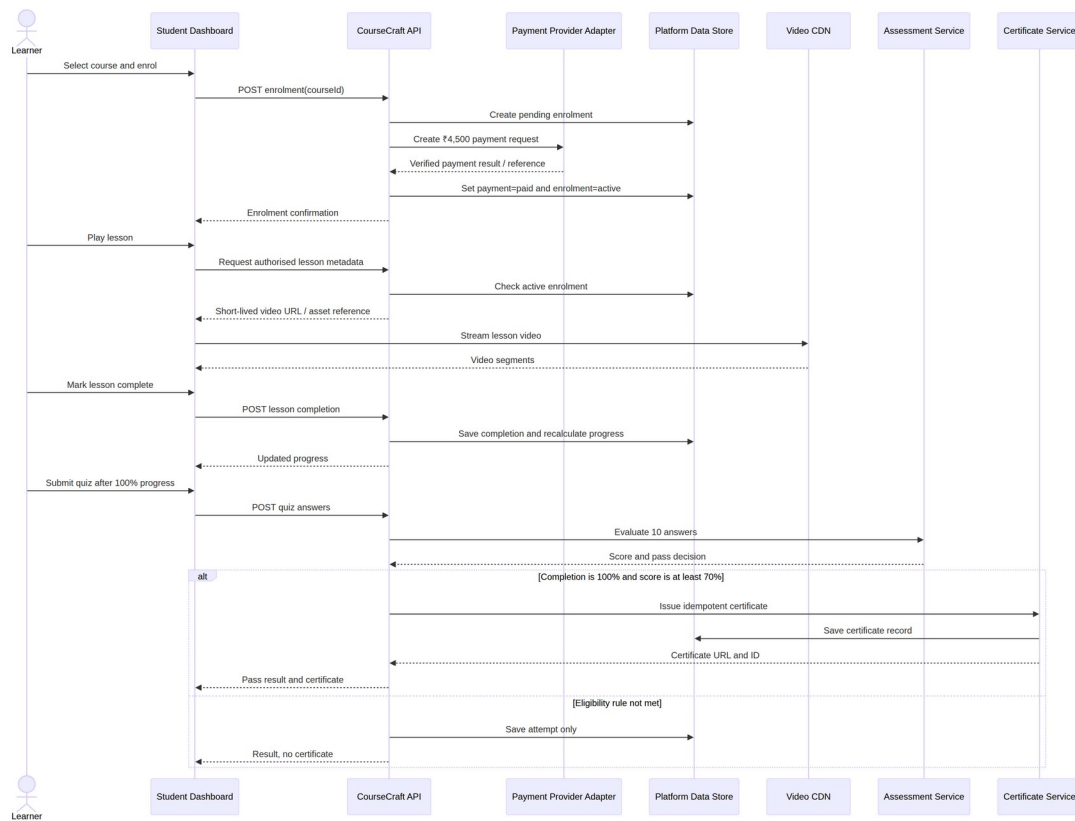
Administrator dashboard use cases

3. Class diagram



Class diagram

4. Sequence diagram — enrol, watch lessons and earn a certificate

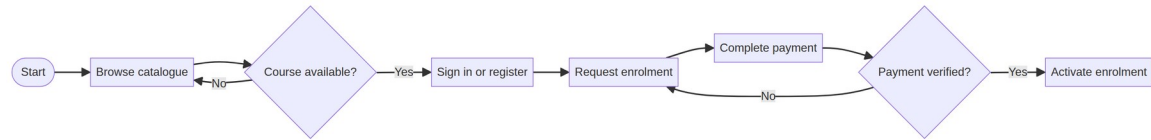


Sequence diagram — enrol, watch lessons and earn a certificate

5. Activity diagram — learner path

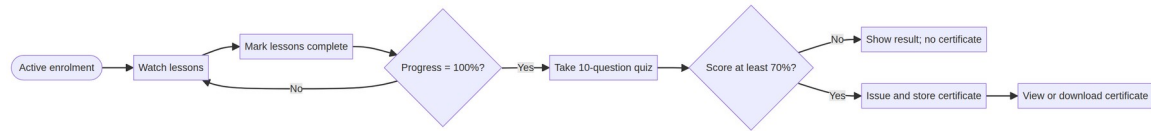
The end-to-end activity is split into two consecutive panels for legibility. Panel B begins after the learner has an active enrolment.

5.1 Enrolment and payment



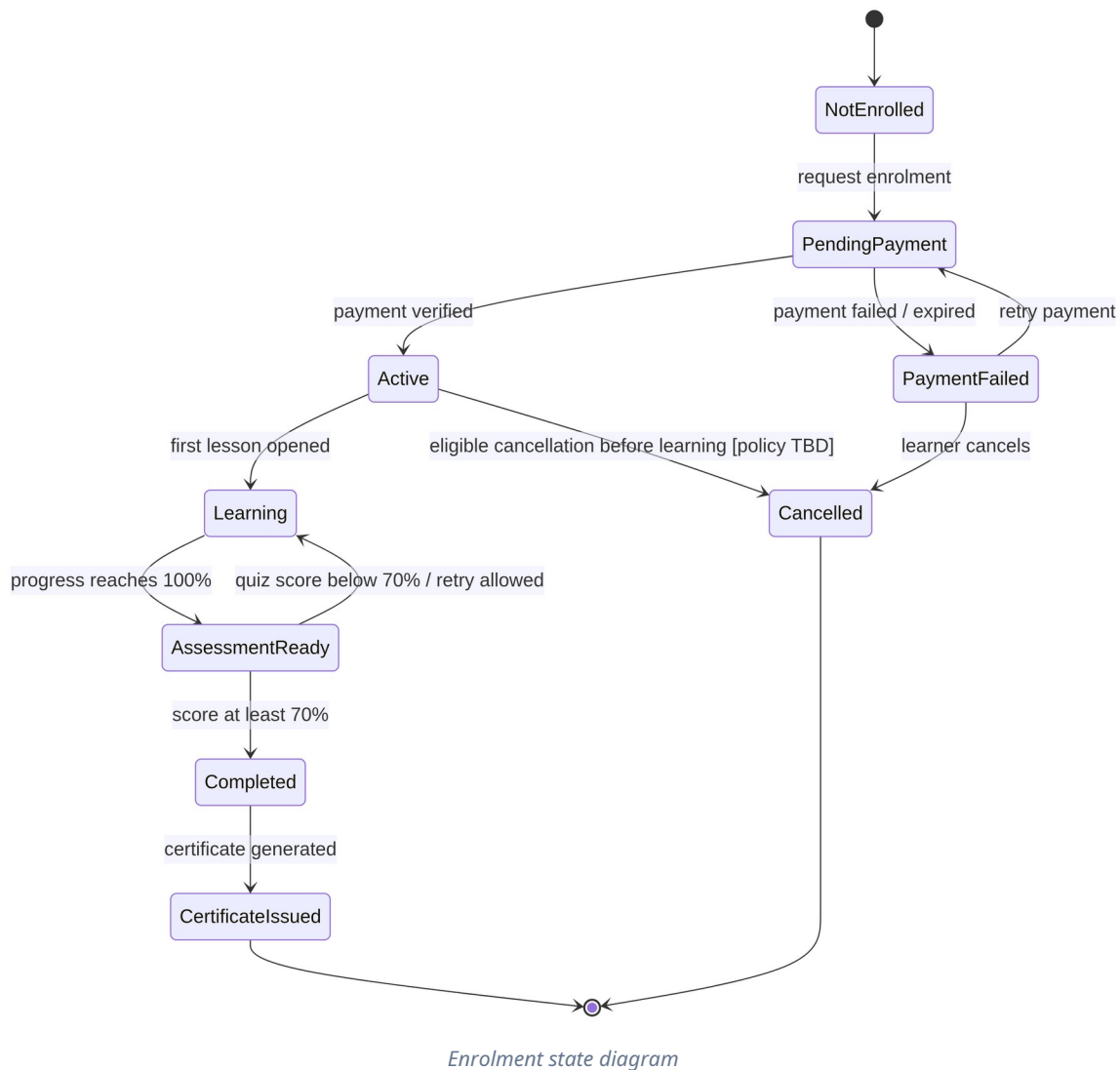
Activity diagram panel A — enrolment and payment

5.2 Learning and certification



Activity diagram panel B — learning and certification

6. Enrolment state diagram



7. Consistency notes

- The sequence and state diagrams both require verified payment before the enrolment becomes active.
- Learning progress is attached to an enrolment, not merely to a user or course, so a learner's record is scoped correctly.
- Assessment eligibility requires 100% progress and a score of at least 70%; the certificate is created only after both conditions hold.
- Payment retry, quiz retry limits, cancellation/refund rules and certificate revocation are intentionally left as policy decisions because the case does not define them.